APPLICATION AND CONTROL OF VARIABLE SPEED CENTRIFUGAL COMPRESSORS IN THE OIL AND GAS INDUSTRY

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Abstract- Centrifugal Compressors in the Oil and Gas Industry frequently use electric motors as drivers. Due to the high variability in process conditions users often specify variable speed electric motors, which allow the variation of the compressor speed over a wide range. The paper discusses the interaction between a centrifugal compressor and the process, and as a result, the control requirements for centrifugal compressor packages. The focus is on variable speed, upstream and midstream applications. The impact of the interaction between system characteristics and compressor characteristics, both under steady state and transient conditions, is explained. Also considered are concepts to optimize and control the units. Special attention is given to the specific speed-torque characteristics of variable frequency drive electric motors and variable speed hydrodynamic drives. Additionally, the impact of the process and how the process dynamics interact with the compressor is analyzed, categorized, and explained.

Index terms: Electric Motors, Centrifugal Compressors , VFD (Variable Frequency Drive), ASD (Adjustable Speed Drive)

I. INTRODUCTION

This paper discusses variable speed centrifugal compressors driven by electric motors. We will briefly discuss the general behavior of variable speed capable compressor trains, and the electric driver technologies used to allow for speed variation. The reason why speed variation is a preferred means of compressor control is outlined in Figure 1: Speed control allows for a drastic reduction in power consumption when the compressor operates at reduced flow. The assumption in Fig. 1 is, that flow and compressor head are related (this is true for many, but not all applications), such that a reduction in flow coincides with a reduction in head [1].

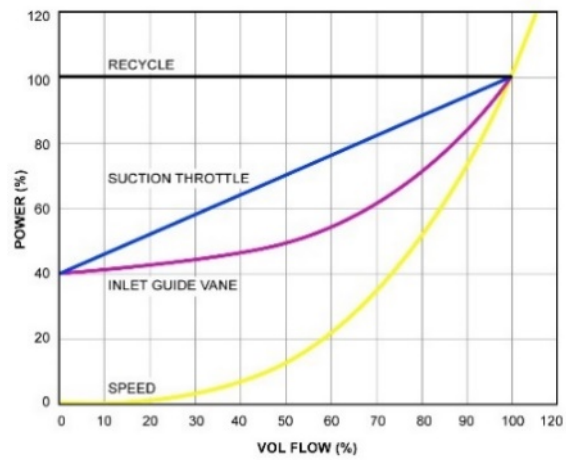

Fig. 1 Power consumption for different control methods

II. EQUIPMENT

A) *The Compressor*

The duty of a gas compressor is to increase the gas pressure, thus reducing the volume of the gas. The faster a compressor runs, and the more impellers it has, the higher the pressure ratio for a given gas.

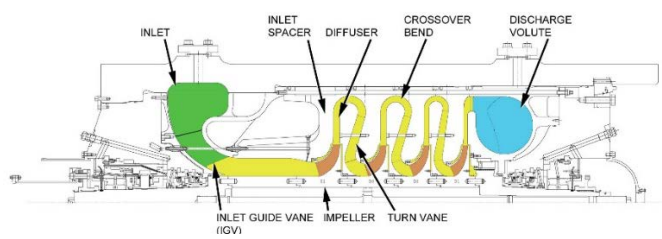

Fig. 2 Cross section of a typical modern barrel type compressor with multiple stages and 2 sections

All centrifugal compressors have one or more stages (Figure 2) Each stage consists of vanes, an impeller, and a diffuser. Vanes are designed to feed the gas flow uniformly into the impeller. A first stage has *inlet* vanes: following stages have *return* vanes. The impeller, the only rotating part of the stage, transfers the mechanical work fed into the compressor into the

gas, and in this process increases the pressure, the temperature and the velocity of the gas. Because the desired outcome of all this work is a pressure increase, the diffuser converts most of the velocity, or kinetic energy, of the gas into pressure. The last diffuser in a compressor often feeds into a volute, which can be thought of as a diffuser in the form of a scroll, and it also converts kinetic energy of the gas into pressure.

The energy content of the gas is expressed as its enthalpy. All the mechanical work that is provided at the compressor shaft, except for a small amount lost to friction in the bearings and seals, is converted into an increase in enthalpy. The increased enthalpy manifests itself as an increase in temperature and pressure of the gas. If the composition of the gas is known, thermodynamic state variables such as pressure, density, temperature, enthalpy, entropy, etc., define the state of the gas as soon as two of the variables are known. Knowing the pressure and temperature of the gas at the compressor inlet and discharge fully defines the enthalpy and entropy, as well as the density of the gas at the inlet of the compressor.

In many instances, rotordynamic considerations define the limits of speed, pressure, and number of impellers. Some applications, such as pipeline transmission, are not particularly challenging from a rotordynamic perspective, since the process only requires small pressure ratios. Other applications, such as gas injection at high discharge pressures, are limited by the capability to achieve rotordynamic stability. Designs have to avoid critical speeds, minimize the excitation of the rotor, and seek designs that provide sufficient damping [1,2]. Multiple compressor bodies (or compressors with multiple sections) are typically required if multiple process streams have to be handled, or if the high pressure ratio requires intercooling (Figures 3 and 4).

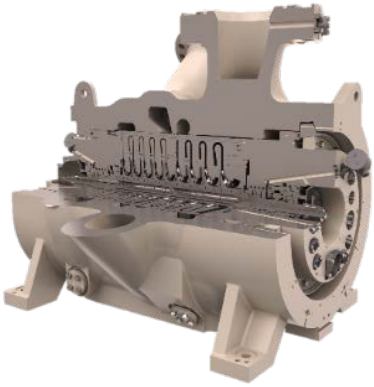

Fig. 3 Compressor with two sections.

Upstream and Midstream applications are dominated by two shaft gas turbine drivers, and variable speed electric motors, with some constant speed electric motor applications. Only constant speed drives rely exclusively on suction throttling or recycling [1], or, for low pressure applications, on adjustable guide vanes. LNG (Liquefied Natural Gas) applications use steam turbines, two shaft gas turbines, single shaft gas turbines and variable speed or constant speed electric motors, as well as turbo expanders as drivers. Downstream and refinery applications are dominated by steam turbines and constant speed electric motors as drivers [2].

A variable speed compressor allows operation over a wide range of head and flow requirements (Figure 5). The actual compressor operating point is the result of the compressor's interaction with the system around the compressor. The system has a distinct relationship between pressure ratio (or head) and flow. Therefore, the system imposes the suction and discharge pressure upon the compressor, which responds with a certain flow. For a variable speed compressor, either the power input from the driver (if it is a gas turbine), or the driver speed (typical for electric motor drives) are used to adapt the compressor to process control objectives.

Fig 4. Multi body tandem

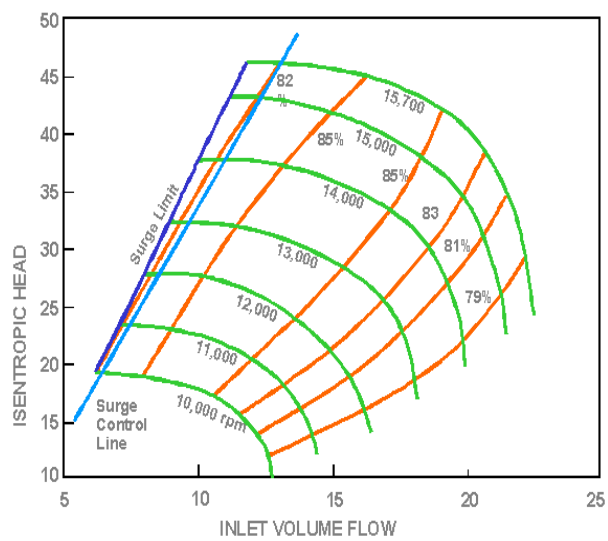

Fig. 5 Typical Compressor Map for a variable speed centrifugal compressor

B) Drivers

For the turbomachinery employed in oil and gas applications, a number of drivers allow for variable speed operation: Gas turbines, expanders, steam turbines and electric motors. Electric motor configurations include constant speed motors driving the compressor via a variable speed gearbox (VSHD, Figure 6) and variable frequency drive speed (VFD) -controlled motors driving the compressor either directly or via a gearbox (Figure 7).

Fig. 6 Electric Motor drive Package, constant speed motor driving the compressor via a variable speed gearbox

Fig. 7 Electric Motor drive Package, VFD controlled motor driving the compressor via a speed increasing gearbox

Since the electric motor speed (for induction type motors, synchronous motors and permanent magnet motors alike) is determined by the frequency of the electric current and voltage supplied to the motor, a convenient way to vary the motor speed is to modify the power supply frequency by using a variable frequency drive [3,4].

Another way to create a variable speed drive is by using a constant speed motor, and a variable output speed gearbox [3], often referred to as Variable Speed Hydrodynamic Drive (VSHD). The basic components of a VSHD are a hydrodynamic torque converter coupled with a planetary gear. The planetary gear is designed as a superimposing gear, with the torque converter driving one of planetary gear components. The torque converter acts as the control unit for the output speed. The VSHD uses a constant speed electric motor. Both the VHSD and the VFD are controlled by setting their speed, until the available power becomes a limit. For the purpose of this discussion we use adjustable speed drive (ASD) to describe various methods to adjust the speed of a variable speed compressor, such as variable frequency drives (VFD) and Variable Speed Hydrodynamic Drives (VSHD)

Of importance for many applications are the performance characteristics of the driver, for example the power as function of ambient conditions, or the power output at various output speeds. In general, a VFD controlled motor is a constant torque machine, thus exhibiting a linear drop in power with speed (Figure 8), implemented by maintaining a constant Volts/Hz ratio, until, above a certain corner frequency, the motor becomes power limited. There are exceptions to this behavior, where the motor is oversized to provide constant power over a wider range (Expanded Power Range, EPR), or where, often for thermal reasons, the torque is reduced with speed. The speed-power relationship has a significant impact on control concepts for variable speed drives. In particular the linear reductions of power seen in most VFD controlled electric motors imposes a limit on the flexibility, compared to VSHD and two shaft gas turbine drives.

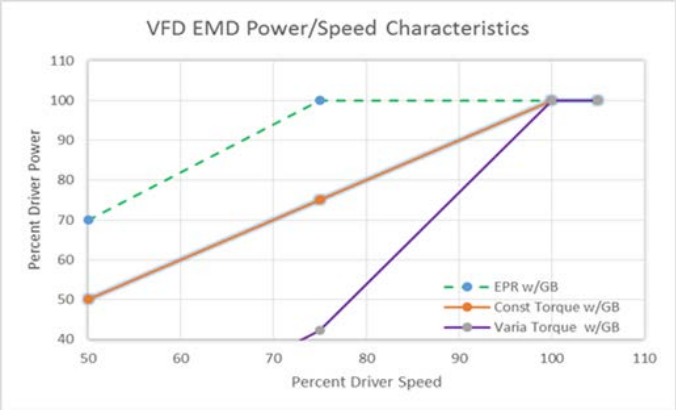

Fig. 8 Speed-power characteristics for variable frequency drives [4].

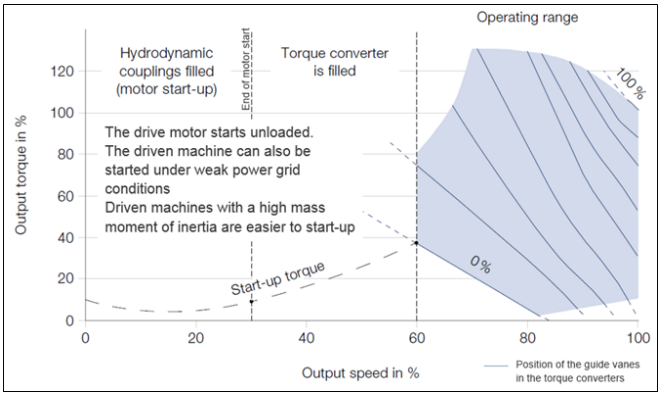

Fig. 9 Operating map of a VSHD [4]

A VSHD shows a speed -power relationship similar to a two-shaft gas turbine, but with a smaller usable speed range (Figure 9).

Another feature of importance is the fact that the electric motor output is not subject to changes in ambient temperature (within limits). This means that it lacks the convenience of providing more power at lower ambient temperatures, but it has the advantage that the same power is available on hot and cold days. This can be important in applications where the load demand is dependent on ambient conditions.

Further, the starting characteristics, including the amount of torque at low speeds, or, for constant speed electric motors, the amount of additional current is required during starting must be considered (Figure 10).

Whether the units are installed on shore, offshore or subsea, determines access to maintenance intervention, as well as the environmental conditions (for example salt in the air) the equipment has to be designed for. For electric drivers, the question is also whether the electricity can be brought to site via transmission lines, or whether it has to be generated on site, usually with gas turbine driven generators.

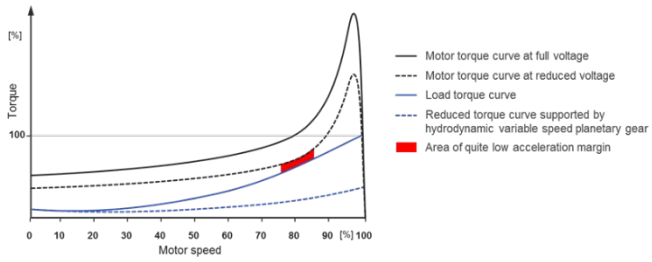

Fig. 10 Squirrel cage induction motor capability curve at different voltage levels and load torque curve [4]

2.3 Packages

Compressors and drivers are usually combined into a package on a skid, which contains the driver, the compressor train, possibly a gearbox, and the necessary couplings (Figures 6). Depending on the configuration, it may include a lube oil system, bearing systems, a control system, sealing systems, air filtration and cooling systems, and an enclosure. Special requirements apply to offshore applications. Platforms in deeper water impose special requirements, since floating platforms dynamically move [2].

III. Oil and Gas Applications

3.1 Applications

Natural gas and oil are found in reservoirs all over the world. The oil and gas are trapped in the pores of the reservoir rock. Reservoirs usually contain mixtures of lighter and heavier hydrocarbons, but also CO_2, Nitrogen, water, and sometimes H_2S. A key feature of a reservoir is its permeability, determining how freely oil and gas can move. Reservoirs with low permeability, restricting the flow of oil and gas will require special techniques to move the oil or gas from the pores to a producing well [2].

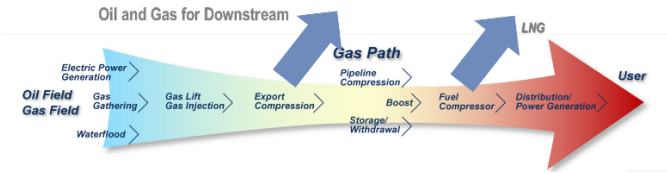

Fig. 11 Natural gas from the well to the user

Even today, with advanced technologies, in some reservoirs more than two-thirds of the oil present may not be recoverable. Once an oil or gas reservoir is discovered and assessed, the task is to bring a sellable product to a user, as well as to maximize the amount of oil or gas that can ultimately be recovered. In the past, natural gas was often seen as a byproduct of the oil production, and may have been flared, especially if there was no infrastructure to bring the gas to potential users. However, even in situations like that, the gas can be used to enhance the recovery of oil. Where the as has value, countries like the USA have built large networks of pipelines to bring gas from the well to users [2,5].

Figure 11 shows the path of the gas from well to user. In general, activities close to the oil and gas field are considered upstream, while activities involving the transport and storage of gas are considered midstream applications [1].

Fig. 12 Offshore platform

Many oil and gas wells are on the ocean floor, and production requires an offshore platform (Figure 12) or subsea installations.

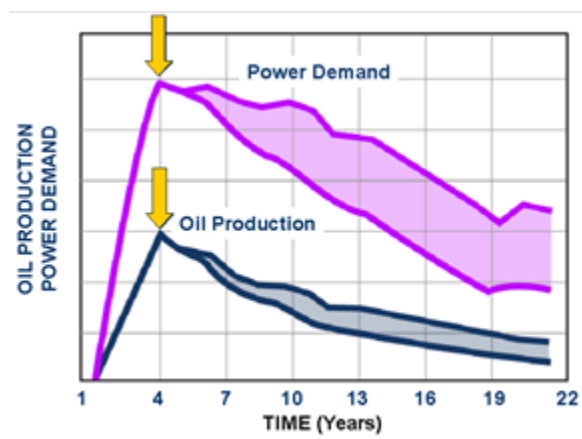

Fig. 13 Typical change in oil production for an offshore well

The reservoirs are at elevated pressure. A series of valves and equipment ("Christmas Tree") are installed on top of the well, to regulate the flow of hydrocarbons out of the well. Early in its production life, the underground pressure will often push the hydrocarbons all the way up the well bore to the surface. Depending on reservoir conditions, this "natural flow" may continue for many years. When the pressure differential is

insufficient for the oil to flow naturally, pumps must be used to bring the oil to the surface. This process is referred to as artificial lift [6].

Most wells follow a predictable pattern where production will increase for a short period, then peak and follow a long, slow decline (Figure 13). The shape of this decline curve, how high the production peaks, and the length of the decline are all driven by reservoir conditions. The decline curve can be influenced by cleaning out the well bore to help oil or gas move more easily to the surface, by fracturing or treating the reservoir rock with acid around the bottom of the well bore to create better pathways for the oil and gas to move through the subsurface to the producing well or by drilling additional wells or by employing enhanced oil recovery (EOR) techniques. The techniques collectively referred to as enhanced oil recovery (EOR) employ, depending on reservoir conditions, water injection ('water flooding') or the injection of various other substances (Hydrocarbons, steam, nitrogen, carbon dioxide) into the reservoir to remove more oil from the pore spaces to increase production [7]. Fracking enhances the pathways between the pores and the well, allowing to economically produce hydrocarbons in formations with poor permeability, for example in shales.

Most oil wells produce oil, gas, and water. This mixture is separated at the surface. Initially, the oil well may produce mostly oil with a small amount of water. Over time, the percentage of water increases. This produced water varies in quality from very briny to relatively fresh. Where this water cannot be used for other purposes, it may be reinjected into the reservoir — either as part of a waterflooding project or for disposal (returning it to the subsurface).

In general, for applications at or near wellheads, the inlet conditions into the compressor, in terms of temperature, pressure and gas composition not only change over time, but also tend to be uncertain at the time when the equipment is sized. For this reason, the attempt to use constant speed drives may lead to significantly over- (or under-) sized equipment.

The oil is then sent to an oil treatment plant. There it is processed in a gas-oil separation system where its pressure is reduced in several stages. In each decompression stage the associated gas (also called flash gas) is released in a separator until the pressure is ultimately reduced to slightly above atmospheric pressure. The crude oil is then sent to a stabilizer column where it is heated and cascaded through a series of bubble trays spaced throughout the column. Hydrogen sulfide (if present) and remaining light hydrocarbons boil off in this process and are collected at the top of the column, while the sweetened heavy crude is drawn off from the bottom. The stabilized oil is then cooled and stored. The streams collected from the top of the stabilizer unit are treated in accordance with environmental regulations.

Gas from the oil production ('associated gas') or gas from gas fields has to be treated, too. Natural gas wells do not produce just gas but usually some amount of liquid hydrocarbons, called condensate. Natural gas wells also often produce water, but the volumes are much lower than is typical for oil wells. The treatment is done in a gas plant, where contaminants (such as CO_2, H_2S, or Mercury) are removed. Further, the heavier hydrocarbons (like Ethane, Propane or Butane), which are valuable as feed stock for the chemical industry, are removed via a cooling process in cold box. The remaining gas, which is now lighter, with a higher Methane concentration, is also known as residue gas or sales gas. It has to be re-compressed to be fed into a pipeline system. The composition and suction pressure of this gas can change significantly, since, depending on the demand for the heavier hydrocarbons, the plant may not remove some or all ethane ('Ethane reject').

Natural gas is usually transported through pipelines (Figure 14), except in cases where a pipeline cannot economically be built. In that case, the gas can be liquefied (LNG: Liquefied Natural Gas) and transported on a ship or a truck. The liquification process involves cooling the natural gas to very low (-260F) temperatures. The refrigeration processes usually employ centrifugal compressors (Figure 15). The operating conditions for the compressor in a pipeline compressor station are subject to large variation in terms of pressures and flow. Practically all installation with electric motor drives require variable speed capability. The operating conditions for compressor in LNG applications can vary by some margin. If these compressors are driven by electric motors, variable speed motors are preferred, since they facility the starting of the compressors, especially from pressurized hold.

As part of the transportation process in pipelines, gas can be stored in storage facilities, which often use former gas fields, or salt caverns. This allows to balance differences in supply and demand on a seasonal or daily basis (Figure 16).

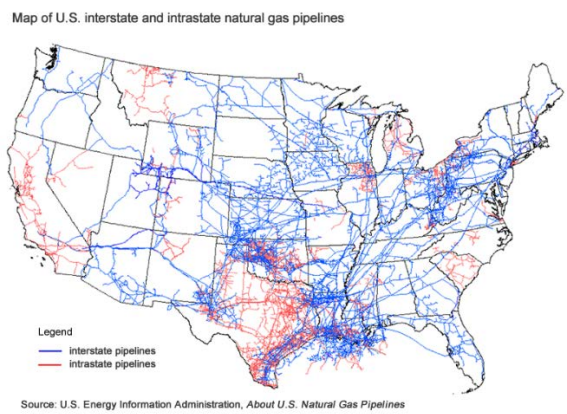

Fig.14 Transmission Pipeline Network [8]

Given the large fluctuations of flow and pressures when the facility is filled, or gas is removed, and fed back into the pipeline, variable speed capable compressor drives are preferred.

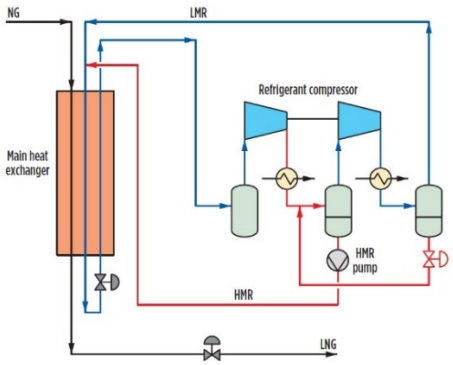

Fig. 15 LNG: Single Mixed Refrigerant Process [Kohler et al.[9]

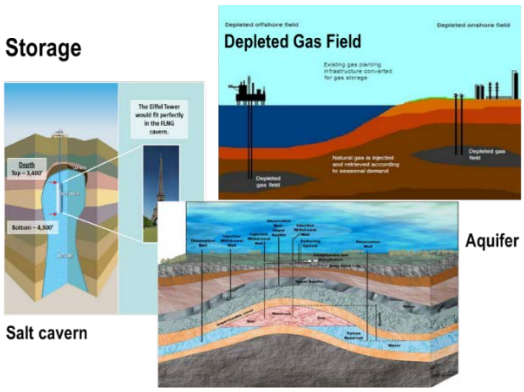

Fig. 16 Gas Storage

Usually, all applications upstream and including a gas plant are considered 'upstream' applications, while the applications related to bringing gas to the ultimate users are referred to as 'midstream'. Applications in refineries, chemical and processing plants are considered 'downstream' applications [2].

In all the applications described above, suction pressure, discharge pressure, gas composition, or flow change over time with the time scales ranging from seconds to years. Being able to change the speed of the compressor allows to adapt to these changes. When sizing VFD driven applications, the steep reduction in power output with speed, compared to two shaft gas turbines or VSHD drives has to be considered. An increase of suction density means that the power consumption for the same operating point in a head flow map (e.g. Figure 5) increases. The drop in output power with speed (Figure 8) causes a reduced capability to accommodate increases in suction density for operating points at lower speeds. It is often assumed that a centrifugal compressor absorbs power according to the fan law, i.e. power reduction is cubed with reduced with speed. This does not apply when the process changes involve changes in inlet density. If the suction density is increased the power demand can increase while the compressor speed is reduced or stays the same.

The gas which has to be compressed usually consists of mixtures of light hydrocarbons (alkanes), nitrogen, and carbon dioxide. In many applications, especially midstream pipeline and storage applications, but also in many upstream applications, the dominant component is Methane. Often, especially in upstream applications near the well, the gas is saturated with water. Hydrogen sulfide may also be present. The gas may also have significant levels of CO_2.In refrigeration applications, heavier hydrocarbons may have to be compressed.

III. COMPRESSOR CONTROL

There are two objectives for compressor control: meeting the external process requirements and keeping the compressor within its operational boundaries. Typical control scenarios that have to be considered are process control, starting and stopping of units, and fast or emergency shutdowns.

Within a compressor station, several compressors may be operated. They can each be driven by a separate motor, or a train consisting of multiple compressors can be driven by a single motor (Figure 4). The compressors can each cover their own, specific duty, but often compressors operate in parallel from the same headers, or in series.

Most systems show some relationship between pressures, pressure ratios, or head, and flow (Figure 17). Pipelines, for example, will require a higher station pressure ratio when the pipeline flow is increased, to compensate for the higher pressure losses in the pipe (Figure 17, Curve A). Applications where the suction pressure is maintained by a process separator, and the gas is fed into a large pipeline will show very little change in pressure ratio with changes in flow (Figure 17, Curve B). Gas Storage applications often run a maximum available power, with a low pressure ratio initially, and a high pressure ratio when the storage gets filled. Since the power is constant, the flow is initially high, and drops later (Figure 17, Curve C). Also, many applications see a change in inlet gas density. As a rule, the system imposes the suction and discharge pressure on the compressor, and the compressor responds with a certain amount of flow into the system.

Control systems have two fundamental functions:

1- Keep the operating parameters of the equipment within safe parameters
2- Maintain process variables.

Process control generally means that a certain process variable (flow, suction or discharge pressure, or power input) is maintained. Importantly, each control parameter that needs to be maintained requires a control method. Control systems also need to keep the compressor within acceptable, safe boundaries. They need to prevent the compressor from surge, running too fast or too slow, at an unsafe discharge pressure, or suction pressure, or operation in choke [3, 8].

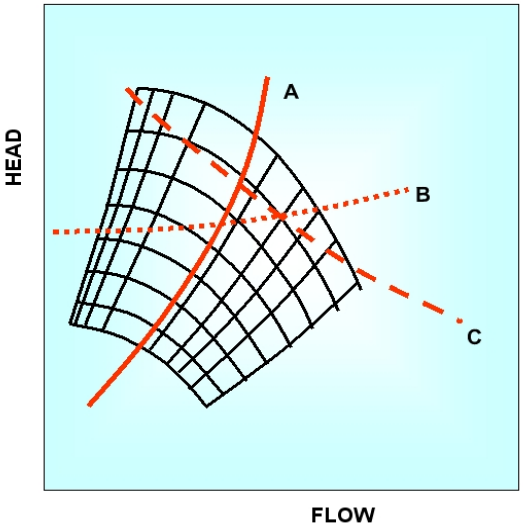

Fig. 17 System characteristics and Compressor Map (schematic)

The following control methods can be used (Figure 18):

1) Power input from driver.
2) Driver speed
3) Guide vane settings
4) Compressor suction or discharge pressure set by a throttle
5) Compressor throughput modified by recycling gas
6) Multiple Units

To prevent the compressor from surging (Figure 5), recycling of gas can be used. Virtually all modern compressor installations use a recycle line with a control valve that allows the flow through the compressor to increase if it comes near the stability limit. The control system determines the distance of the operating point from the surge line, and automatically opens the recycle valve gradually to keep the compressor operating point from crossing the surge limit to lower flows. This avoids surge, a flow reversal in the compressor, which can damage the compressor. For most applications, the operating mode with open, or partially open recycle valve is only used for start-up and shut-down, or for brief periods during upset operating conditions. If this recycle loop is cooled, it allows continuous operation of the compressor station at a flow turndown to very low or no flow without having to shut the compressor down.

Centrifugal Compressors, when driven by drivers capable of variable output speed, are usually adapted to varying process conditions by means of speed control. This is the most natural way of controlling a system, because the centrifugal compressor can operate over a wide range of speeds without any adverse effects. A typical configuration can operate down to 50% of its maximum continuous speed, and in many cases even lower. Reaction times are very fast, thus allowing a continuous load following using modern, PLC based controllers. Typical process controls, for example for flow or pressure, use the change in speed to adapt the compressor.
For gas turbine drivers, the variable speed is actually controlled by power input. Electric motor drives are usually controlled by speed, but a change in speed always leads to a change in power demand from the compressor. For any given situation, the maximum speed is limited by the available motor or VFD power. For VFD driven motors, the available power may depend strongly on speed (Figure 8), and Figure 19 illustrates this for a pipeline system. Given a pipeline characteristic, the compressor operating point is at the intersection between the selected speed and said pipeline characteristic, as long as the motor has enough power (Figure 17, Curve A). Otherwise, the speed that can be achieved is limited by the power of the motor.

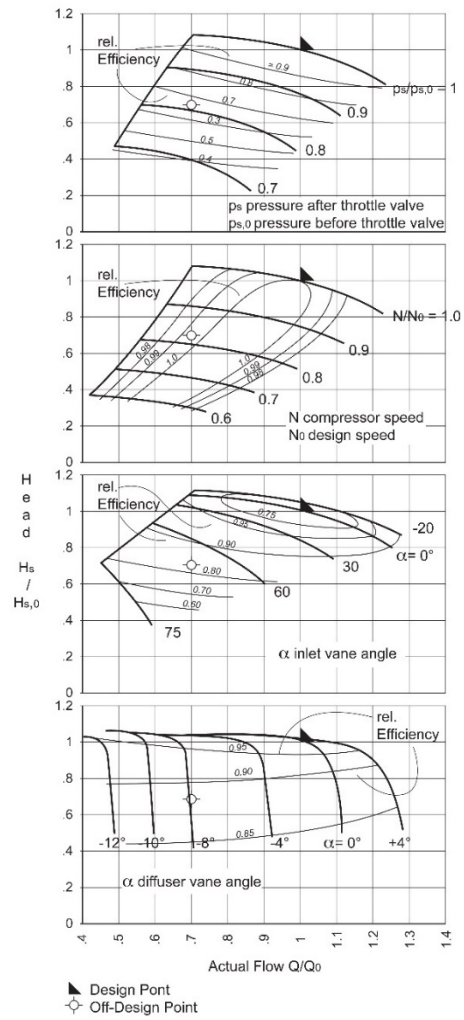

Fig 18: Control methods for centrifugal compressors: Throttling, variable speed and adjustable guide vanes (Rasmussen et al [6]).

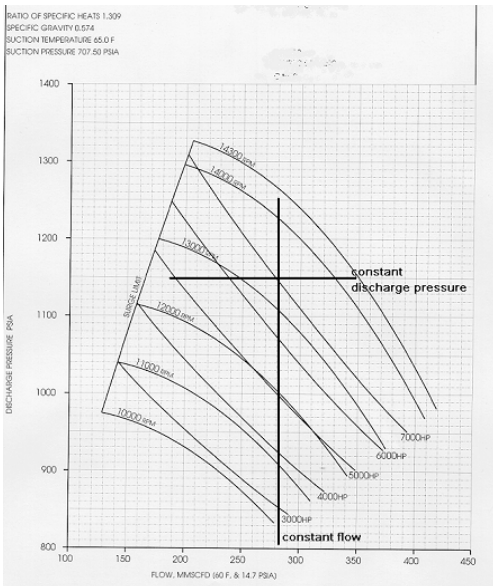

Fig. 19 Control Modes

III. PROCESS CONTROL

As an example for process control, discharge pressure control is described (Figure 19): The discharge pressure is sensed by a transmitter. A pressure set point is selected by the operator. If the discharge pressure decreases, for example because the gas gets lighter, or the suction pressure drops, due to process changes, the controller will increase the speed of the electric motor (or the output speed of a VSHD by changing the converter guide vane angles). Thus, the driver causes the driven compressor to accelerate to maintain the compressor discharge pressure. Both the compressor speed and the power increase in that situation. The action is limited by the available driver power, which can be also a function of speed (Figures 8 and 9). Similar control mechanisms are available to keep the flow constant, or to keep the suction pressure constant. The control scheme works for one or more compressors, and can be set up for machines operating in series as well as in parallel.

In a compressor station with multiple compressors, either identical units or dissimilar units, turning individual compressors on or off provides an effective means of control. Particularly in applications that see large swings in load, this control method allows to operate the compressors near their best efficiency conditions for most of time. The method works well if the selected equipment has a high starting reliability. It is also advantageous if the compressors can stay pressurized and can be restarted without having to vent gas into the atmosphere. Modern centrifugal compressors with dry gas seals are capable of this ‘pressurized hold’. Since the starting of a pressurized compressor requires a significant amount of torque at low speed, the driver capability needs to be evaluated carefully.

Load sharing can be accomplished by either equalizing compressor speed, equalizing the driver power or equalizing the turndown. Turndown is the relative distance of an operating point from the compressor surge line at constant head. For example, if compressors operate in parallel from the same suction and discharge header, all compressors are subject to the same suction and discharge pressure. Thus, changing the speed in unison will change the flow, thus the turndown. If the compressor sets are identical and the recycle valves are identical the task is simplified. However, turndown equalization also works for dissimilar units.

The preferred method of operating identical compressor sets together is turndown equalization. Turndown equalization works equally well for compressor sets in series or parallel. The challenge comes when the compressors reach their respective surge protection margins and begin recycling. At this point the turndown of the compressors is controlled by the surge avoidance system.

At the point recycle begins or ceases to be necessary all compressor sets must switch to the new control mode simultaneously. In recycle, the recycle valve commands are equalized. Then further turndown can be accomplished by reducing compressor speeds equally. If demand increases and recycle ceases to be necessary, they switch back to turndown equalization.

Multiple compressor sets in series may have the same drivers but may not consist of the same compressors nor operate at the same power. Multiple compressor sets in series can be operated at equal turndown, if the high pressure compressor is designed for lower actual flow. Another method of control is to maintain the compressors at their design percentage of the total pressure rise. With this control strategy the surge control systems can be allowed to operate independently. If part of the operational envelope includes recycling, an alternate control strategy must be adopted.

For dissimilar compressors working in parallel, for example one or more large compressor sets in parallel with one or more smaller compressor sets, the larger units would form the baseline compression, as they are often more efficient. The smaller less efficient units may operate as peaking units. Typically the smaller units would be started and shutdown as necessary and recycling would be avoided.

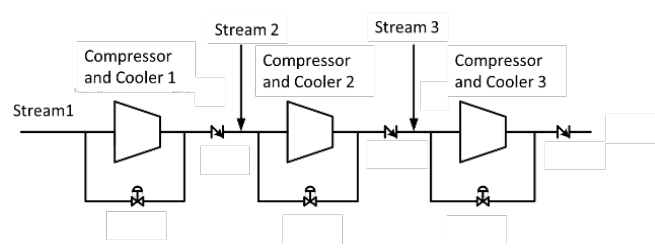

Fig. 20 Compressor train with 3 compressors and two side streams.

Many compressor applications require the simultaneous control of several gas streams (Figure 20), for example where multiple gas streams at different pressure levels have to be compressed, such as in flash gas compressors, or in gas gathering applications where the wells produce at different pressure levels. As a general rule, each independent gas stream needs a separate control mechanism. In these cases, combinations of speed control, suction throttling, or recycling can be used. It is usually advantageous to control the larger

stream via speed control, and the smaller stream via recycle or throttle control, because the speed control is more energy efficient. In a train with multiple compressors, driven by the same motor, where each compressor has to meet certain set points, the control strategy has to address the coordination of the different control modes. Motor speed (and with it motor power) must be allowed to increase until the last compressor achieves its control objective. The other compressors, having passed their set points, are controlled by recycling or throttling.

Process conditions change over various time scales. Therefore, the system behavior under transient conditions has to be considered. A pipeline, for example, is often operated in a transient condition by feeding more gas into the pipeline than what is taken off on the other end. This is usually referred to as line packing. As outlined in Figure 17, a pipeline under steady state conditions causes a unique station pressure ratio for a given flow. Under transient conditions, this is no longer the case.

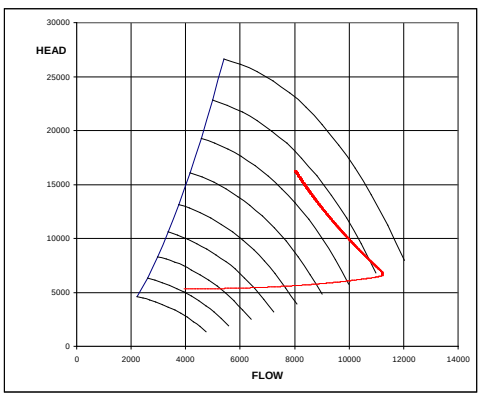

Fig. 21 Trace of operating points for a pipeline compressor accelerating rapidly to full load [10].

Studies of dynamic pipeline behavior reveal a distinctly different reaction of a pipeline to changes in station operating conditions than a steady state calculation. In steady state (or, for slow changes), pipeline hydraulics dictate an increase in station pressure ratio with increased flow, due to the fact that the pipeline pressure losses increase with increased flow through the pipeline (Fig. 17). However, if a centrifugal compressor receives more driver power, or increases its speed then the throughput increases rapidly, while the station pressure ratio will react very slowly to this change (Figure 21). This is due to fact that initially the additional flow has to pack the pipeline (with its considerable volume) until changes in pressure become apparent. Thus, the fast change in operating conditions would lead to a change in flow without a change in head. Eventually, the available driver power limits the compressor speed. If the power setting at maximum power is maintained, the compressor operating point would then start to approach the steady state line again, albeit at a higher speed, pressure ratio, flow and power. In general, changes are slow enough that the steady state compressor maps remain valid. However, for very fast changes, such as they occur during station emergency shutdown conditions, this may only be approximately true (Blieske et al [11], Morini et al [12]).

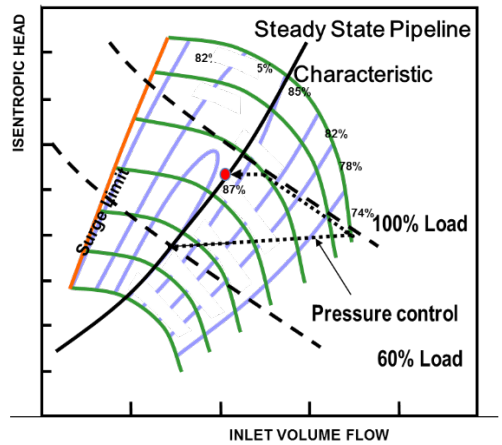

Figure 22: Load Increase. Pressure control [5]

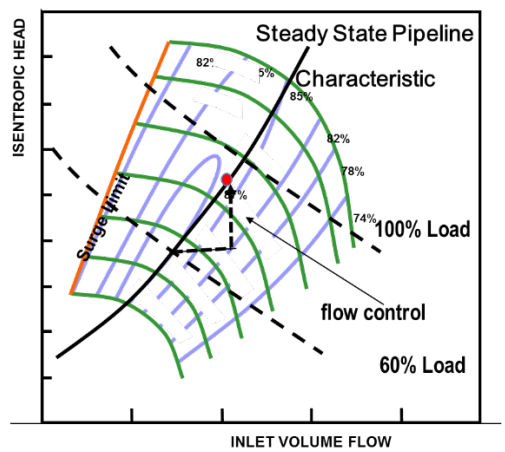

Figure 23: Load Increase. Flow control [5]

As outlined earlier (Figure 19), control systems either control for suction or discharge pressure or for flow. If the motor operates at part load, a new set point, for example for a higher discharge pressure or a higher flow will make the motor increase its speed until the new set point is reached, assuming the motor has sufficient power. Moving to the new operating point will usually be performed at maximum available power. In the case of pressure control, the compressor will, for a period of time operate at low efficiency, in choke (Figure 22). However, the full load, steady operating point will be reached relatively fast because the motor will operate it maximum load most of the time. In the case of flow control, the compressor reaches the flow setpoint (but not the steady state operating point) relatively fast, and then operates at part load. However, because the pipeline imposed pressure ratio increases, the speed, and with it the power demand start to rise until the new steady state operating point is reached (Figure 23). The acceleration could also be performed roughly along the best efficiency area of the compressor, but that would require a much slower acceleration, in other words the maximum motor acceleration would have to be limited. The

transition from the old to the new operating point would in this case be slower, but more efficient.

IV. CONCLUSIONS

Using the capability to operate variable speeds allows the best possible adaptation of centrifugal compressors to the changing operating conditions in typical oil and gas applications. Understanding the behavior of compressors, their drives, as well as the system response allows to find optimum control strategies.

V. VITA

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